

NAT and Port Forwarding

Understanding How Private Networks Communicate with the Internet

Name: Patel Kunjkumar Pinkeshkumar

Enrollment Number: 240760107141

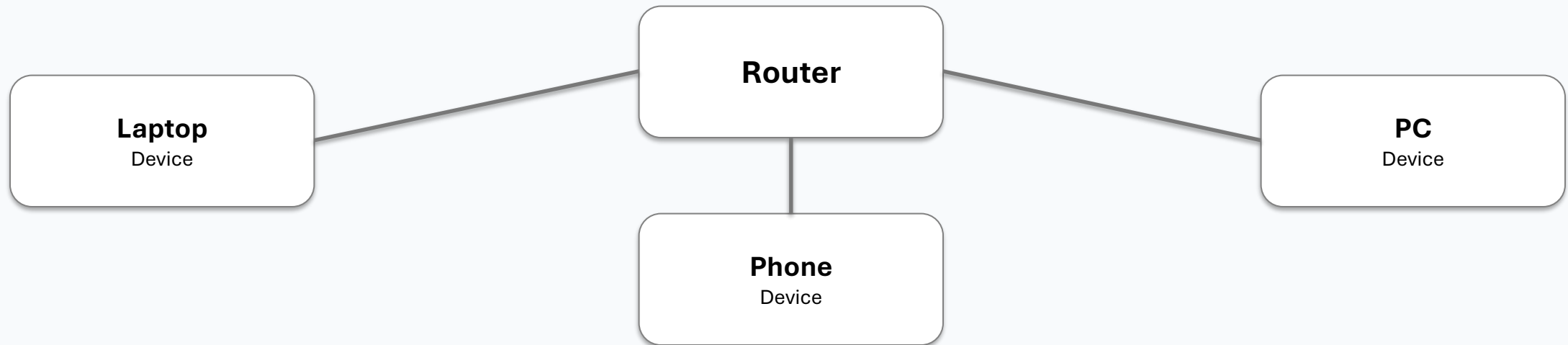
Subject: Computer Networks

Class: Computer 33

What is a Network?

Definition

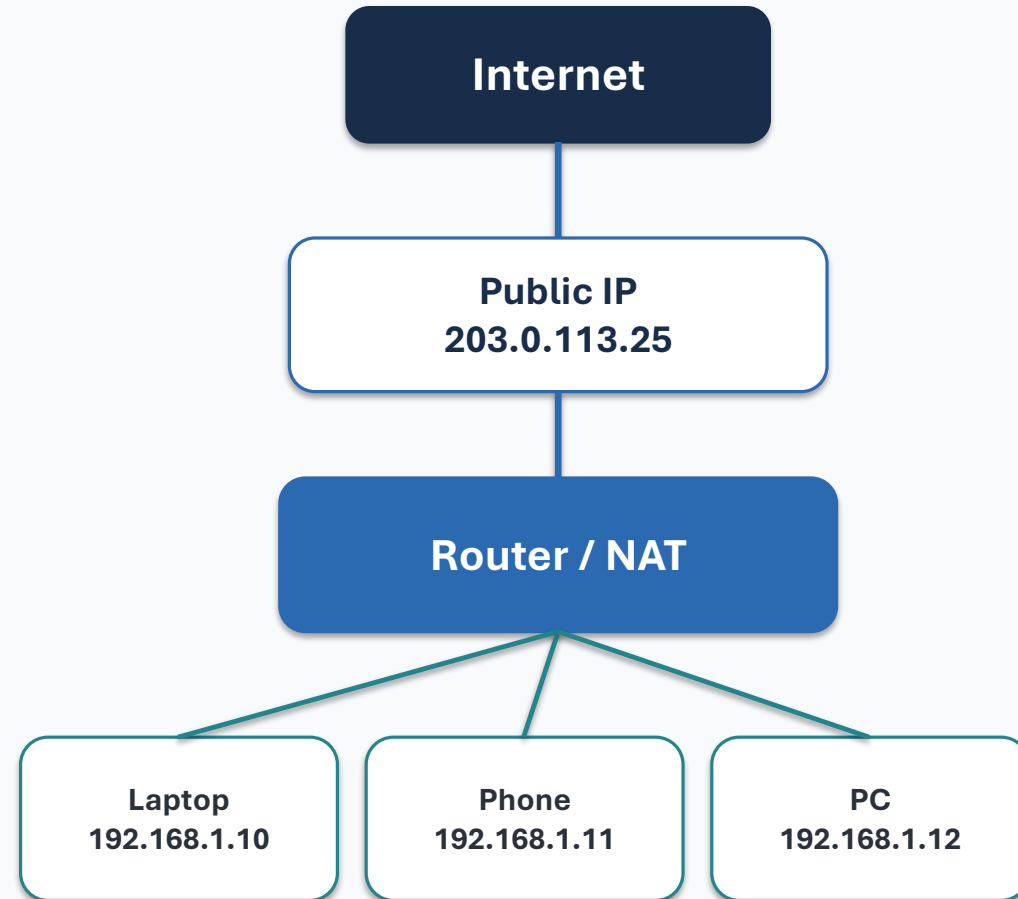
A network is a group of two or more connected devices that can communicate with each other and share data, resources, or services.



Examples: Home Wi-Fi • College Network • Office Network • The Internet

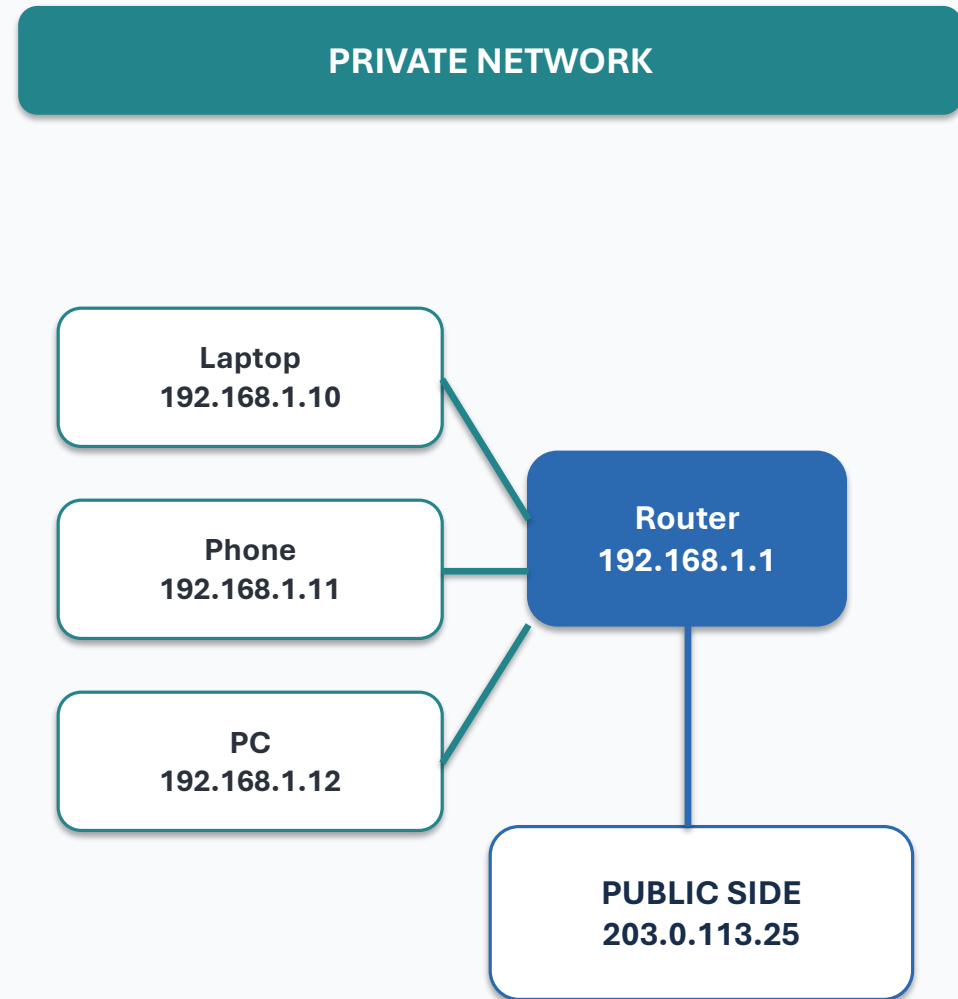
Why Do We Need NAT?

- IPv4 provides a limited number of public IP addresses.
- Homes and organizations may contain many connected devices.
- Private addresses can be reused inside separate local networks.
- NAT lets many private devices share one public IPv4 address.



Private vs Public IP Addresses

- Private IP addresses are used inside local networks.
- They are not directly routable on the public Internet.
- Public IP addresses identify a network/device to the Internet.
- Common private IPv4 ranges: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.



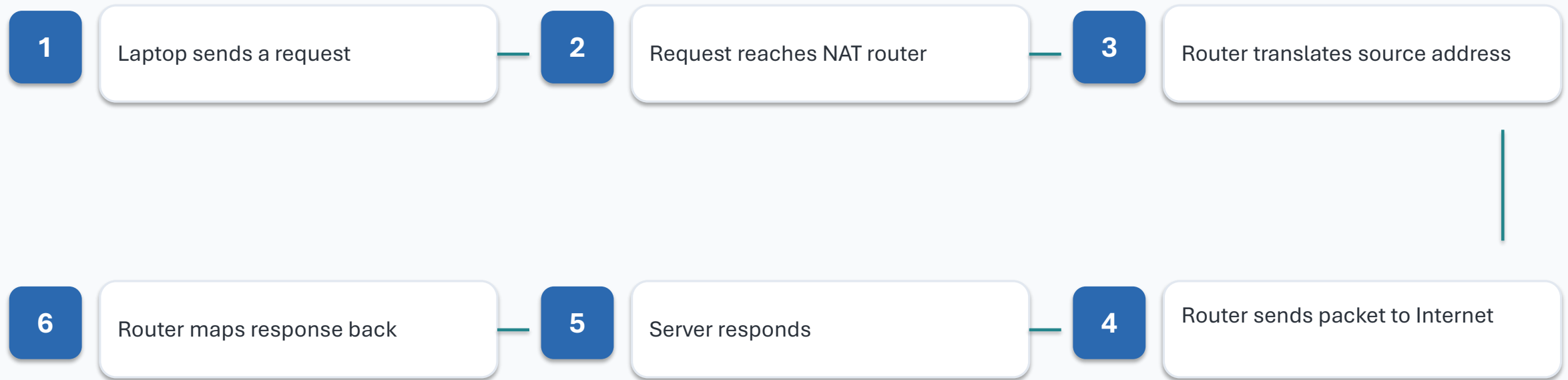
What is NAT?

NAT Network Address Translation

- Translates IP addresses between private and public networks.
- Commonly performed by a router or firewall.
- Allows private devices to access the Internet using a public IP.
- For stateful NAT/PAT, the router maintains translation mappings so returning traffic can be associated with the correct internal connection.



How NAT Works



Outbound request → translated → Internet → response → translated back

Types of NAT

Type	Mapping	Typical Use
Static NAT	One private IP ↔ one public IP	Fixed mapping
Dynamic NAT	Private IPs → public IP pool	Address pool
PAT / NAT Overload	Many private IPs → one public IP + ports	Home networks

PAT is why many devices can share one public IPv4 address simultaneously.

PAT (Port Address Translation) Table

Internal Address	Internal Port	Public Address	Public Port
192.168.1.10	5000	203.0.113.25	40001
192.168.1.11	5001	203.0.113.25	40002

- The translated port numbers allow multiple simultaneous connections to share the same public IP address.
- The router maintains mappings between internal and external connections.
- Multiple devices can therefore share one public IP.
- Ports distinguish simultaneous connections.

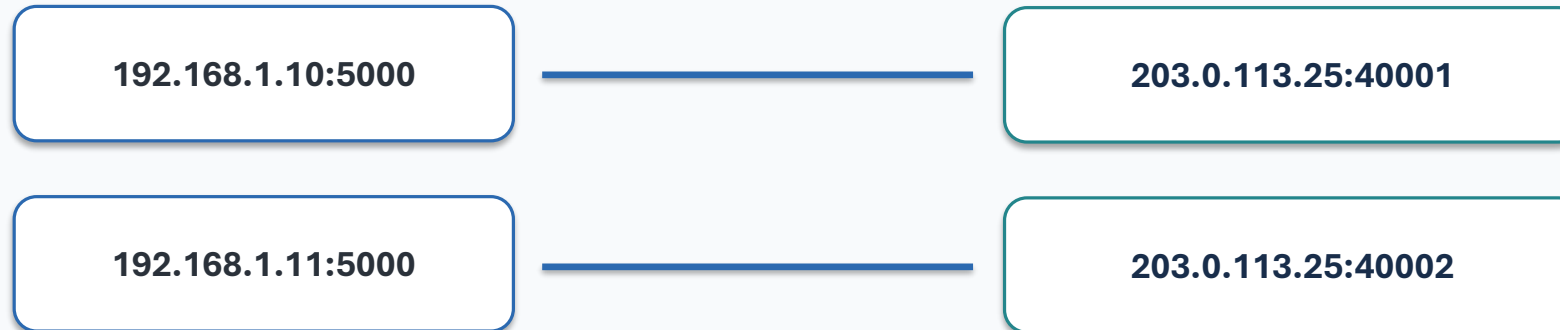
NAT vs PAT

NAT

- NAT is the broader concept of translating addresses. PAT is a type of NAT that also uses transport-layer port numbers, allowing many private devices to share one public IP address.
- Can use one-to-one or pool-based mappings.

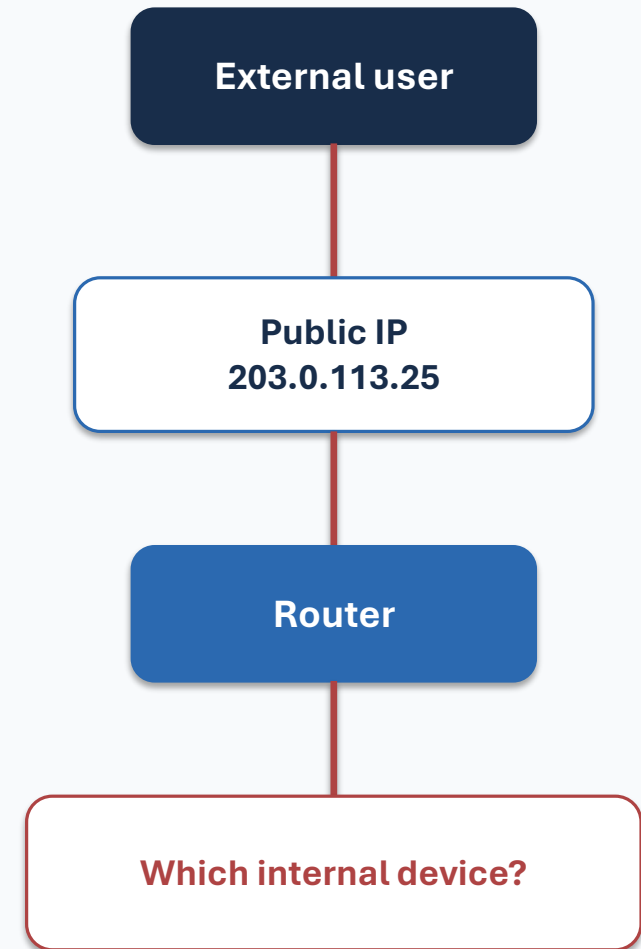
PAT / NAT Overload

- Translates IP addresses AND port numbers.
- Many internal devices share one public IP.
- Also called NAT Overload.



The Problem with Incoming Connections

- Outgoing connections are easy for NAT to track because the internal device starts the connection.
- An Internet user may want to connect to a server inside a private network.
- A private IP address is not globally routable on the public Internet.
- The router needs to know which internal device should receive the connection.

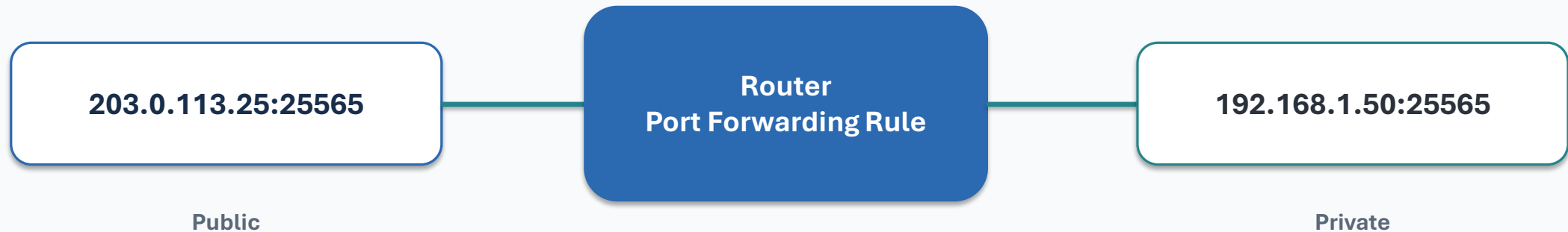


This is where Port Forwarding is used.

What is Port Forwarding?

A router/firewall rule that forwards incoming traffic on a specific public port to a specific internal IP address and port.

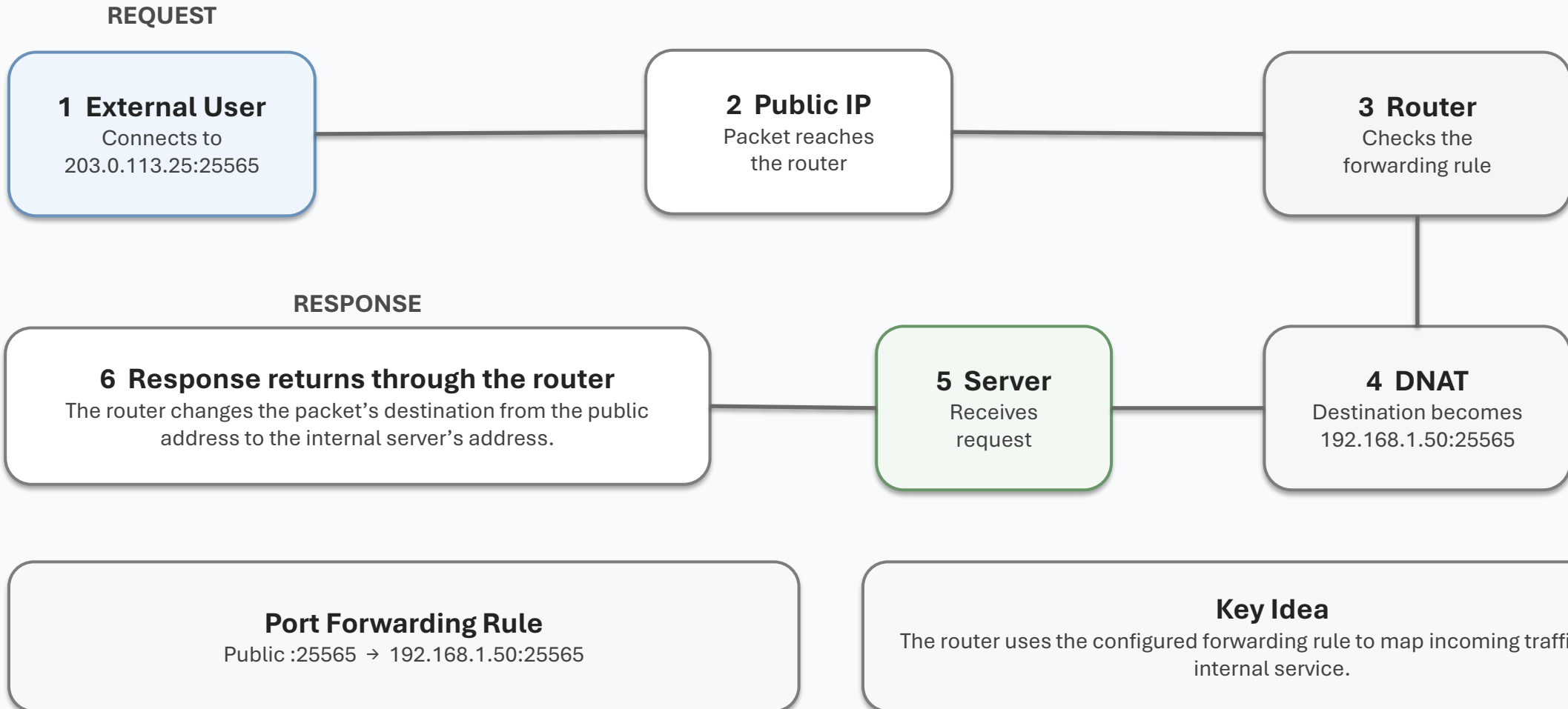
Port forwarding is commonly implemented using Destination NAT (DNAT).



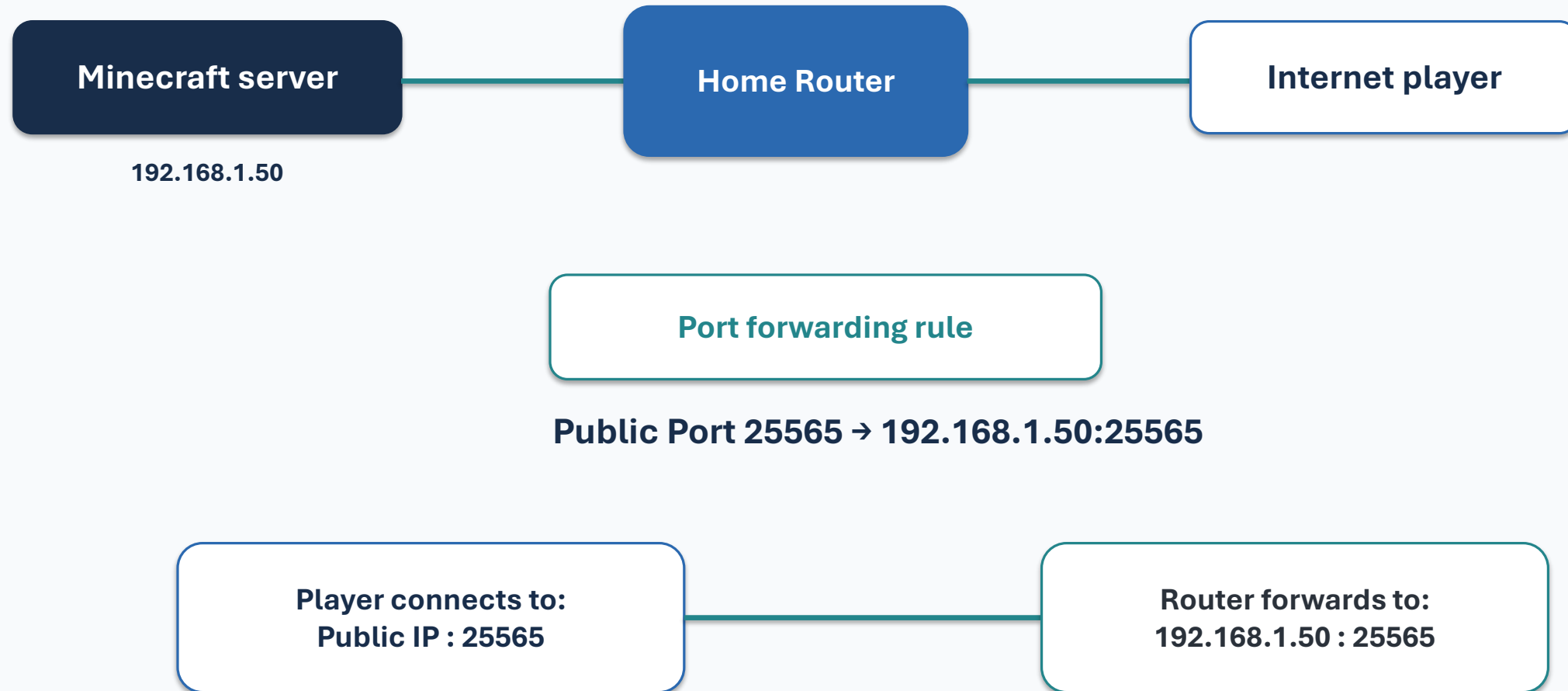
Creates an explicit rule for directing incoming traffic to an internal service.

How Port Forwarding Works

A request travels from the Internet to a specific service inside the private network



Real-World Example: Hosting a Game Server



Other Uses of Port Forwarding

Hosting a game server

Web server

SSH server

Remote access to Internal services/devices

CCTV / NVR systems

Self-hosted applications

Different services use specific ports or port ranges; the router can forward the required external ports to the appropriate internal service.

NAT vs Port Forwarding

NAT	Port Forwarding
Translates addresses between networks	Directs incoming traffic to an internal service
Commonly helps internal devices access the Internet	Commonly enables external access to an internal service
Often supports outbound connections automatically	Requires an explicit rule
Primarily about address translation	Primarily about directing incoming traffic to a specified internal IP and port.



Security Considerations

- Opening a port makes a service reachable from the Internet.
- Only forward ports that are actually required.
- Keep the service and operating system updated.
- Use strong authentication.
- Avoid exposing unnecessary administrative services.
- Use firewalls and access controls.
- Monitor exposed services.
- Prefer secure protocols such as SSH instead of Telnet.

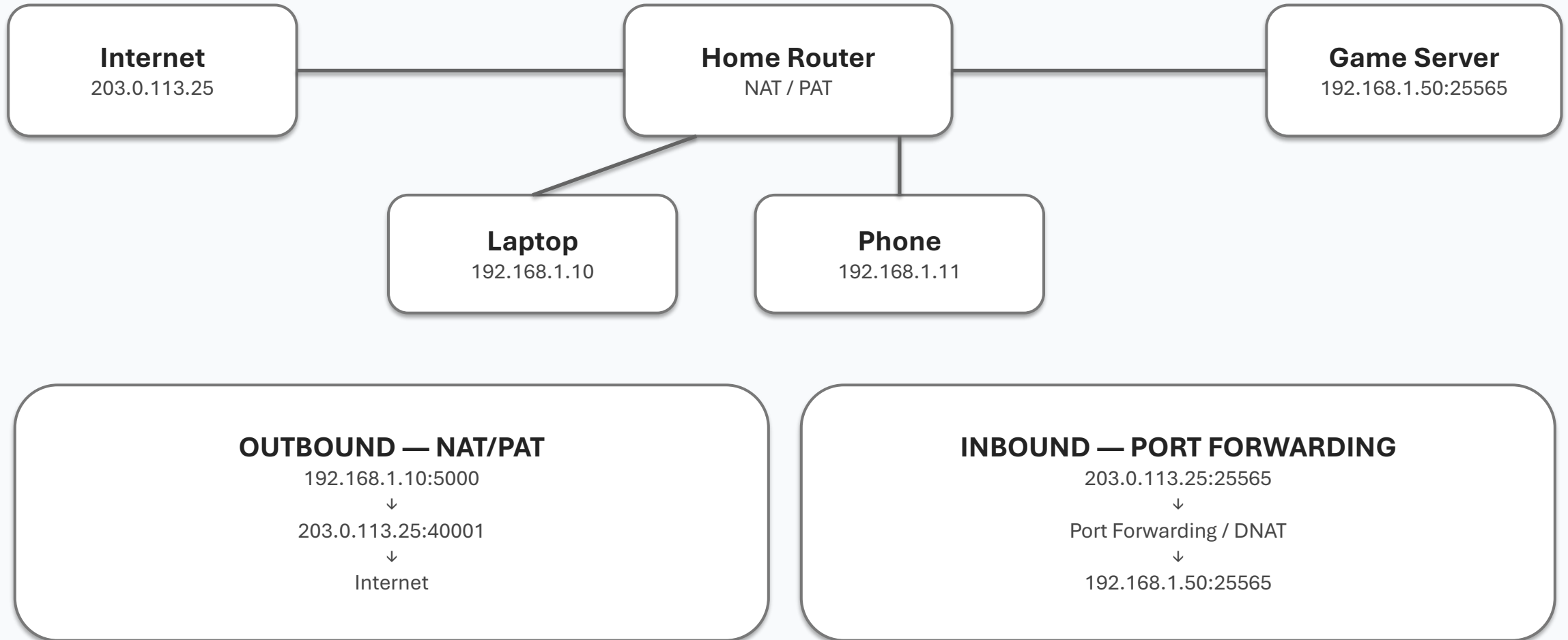
Important

Port forwarding itself is not a security vulnerability.

Port forwarding is not inherently a vulnerability, but it increases exposure to Internet-based attacks if the forwarded service is vulnerable or poorly secured.

Complete Example: NAT/PAT + Port Forwarding

One router handles outbound sharing and selected inbound access



NAT/PAT handles outbound communication; port forwarding enables selected inbound communication.

THANK YOU